

## CSA1304-THEORYOF COMPUTATION LAB EXPERIMENTS

9. Write a C program to simulate a Non-Deterministic Finite Automata (NFA) for the given language representing strings that start with 0 and end with 1

### AIM

Write a C program to simulate a **Non-Deterministic Finite Automata (NFA)** for the language representing strings that **start with 0 and end with 1**.

### CODE

```
#include <stdio.h>
#include <string.h>

int main()
{
    char str[100];
    int len;

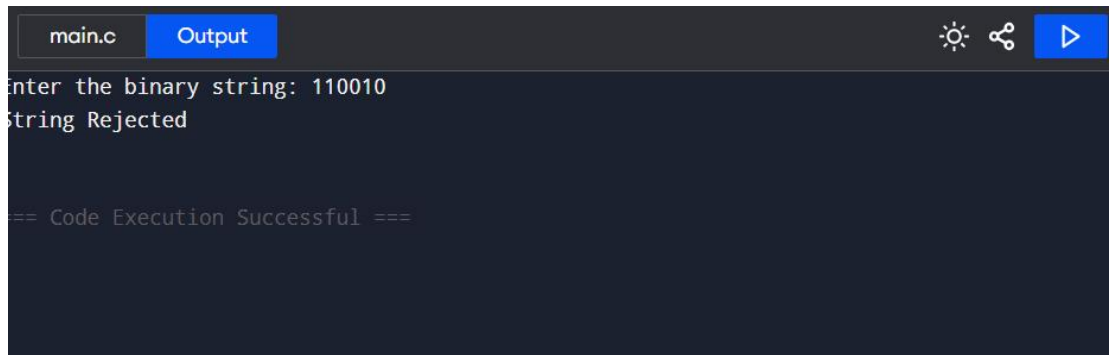
    printf("Enter the binary string: ");
    scanf("%s", str);

    len = strlen(str);

    if(len > 0 && str[0] == '0' && str[len - 1] == '1')
        printf("String Accepted\n");
    else
        printf("String Rejected\n");

    return 0;
}
```

## OUTPUT



The screenshot shows a code editor interface with a dark theme. At the top, there are two tabs: 'main.c' and 'Output'. The 'Output' tab is active, displaying the program's output. The output text is as follows:

```
Enter the binary string: 110010
String Rejected

== Code Execution Successful ==
```

The output window also features icons for settings, sharing, and a play button in the top right corner.